Regulation and Pharmacology of the Hypothalamus and Pituitary

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No conflicts of interest to disclose

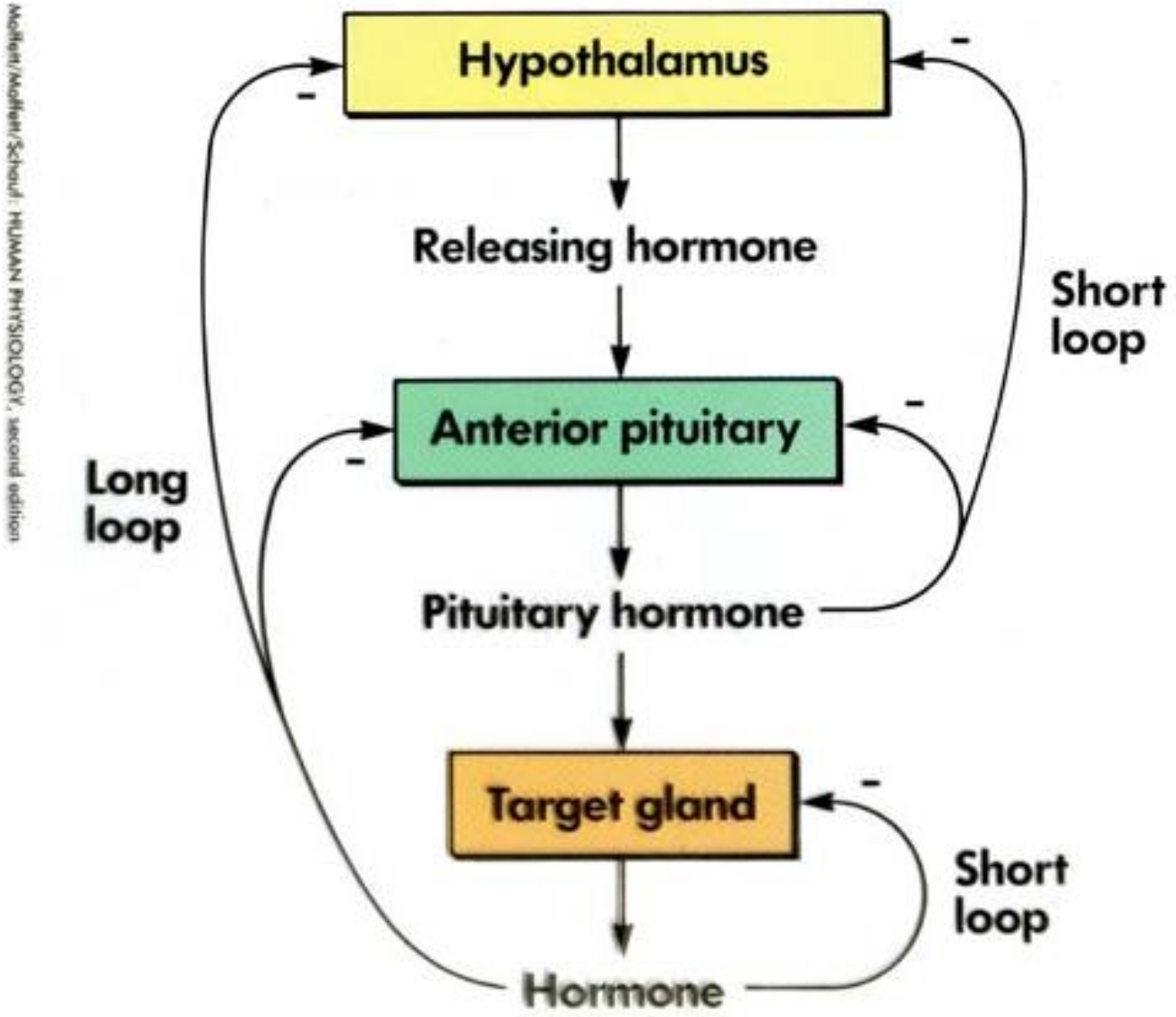
Learning Objectives

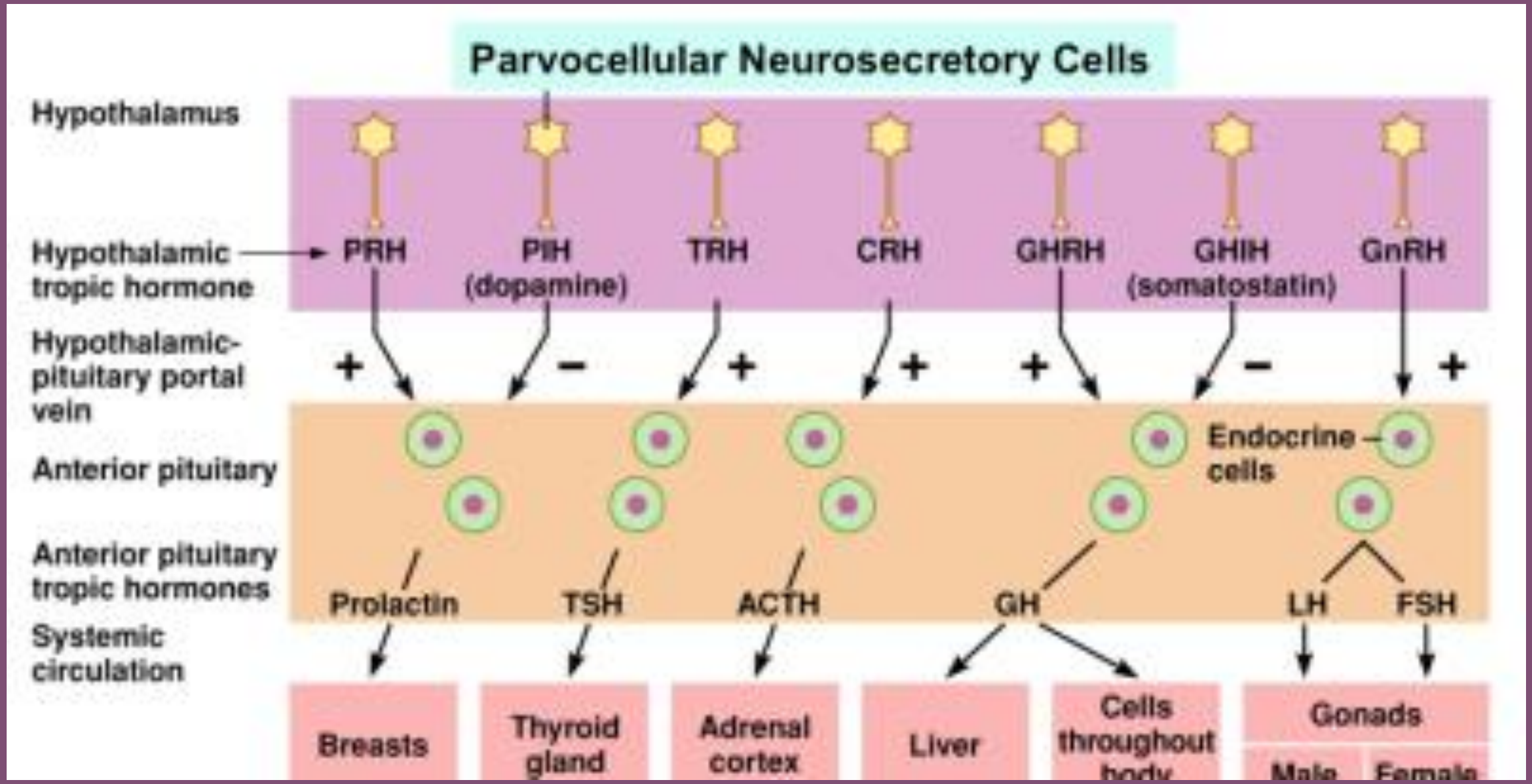
1. Recall the hormones, target organs, functions and feedback regulation of the hypothalamic-pituitary-growth hormone axis – our model for understanding the system. PHYS
2. Explain the pharmacologic treatment of growth hormone excess and deficiency, including the mechanisms of action, key adverse effects, and monitoring requirements of relevant medications. PHARM
3. Describe the hormones, target organs, functions, and feedback regulation of the hypothalamic-pituitary-adrenal axis. PHYS

4. Explain the pharmacologic treatment of adrenal hormone excess (primary and secondary) and deficiency (primary and secondary), including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM
5. Describe the hormones, target organs, functions, and feedback regulation of the hypothalamic-pituitary-thyroid axis. PHYS
6. Explain the pharmacologic treatment of thyroid hormone excess (primary and secondary) and deficiency (primary and secondary), including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM

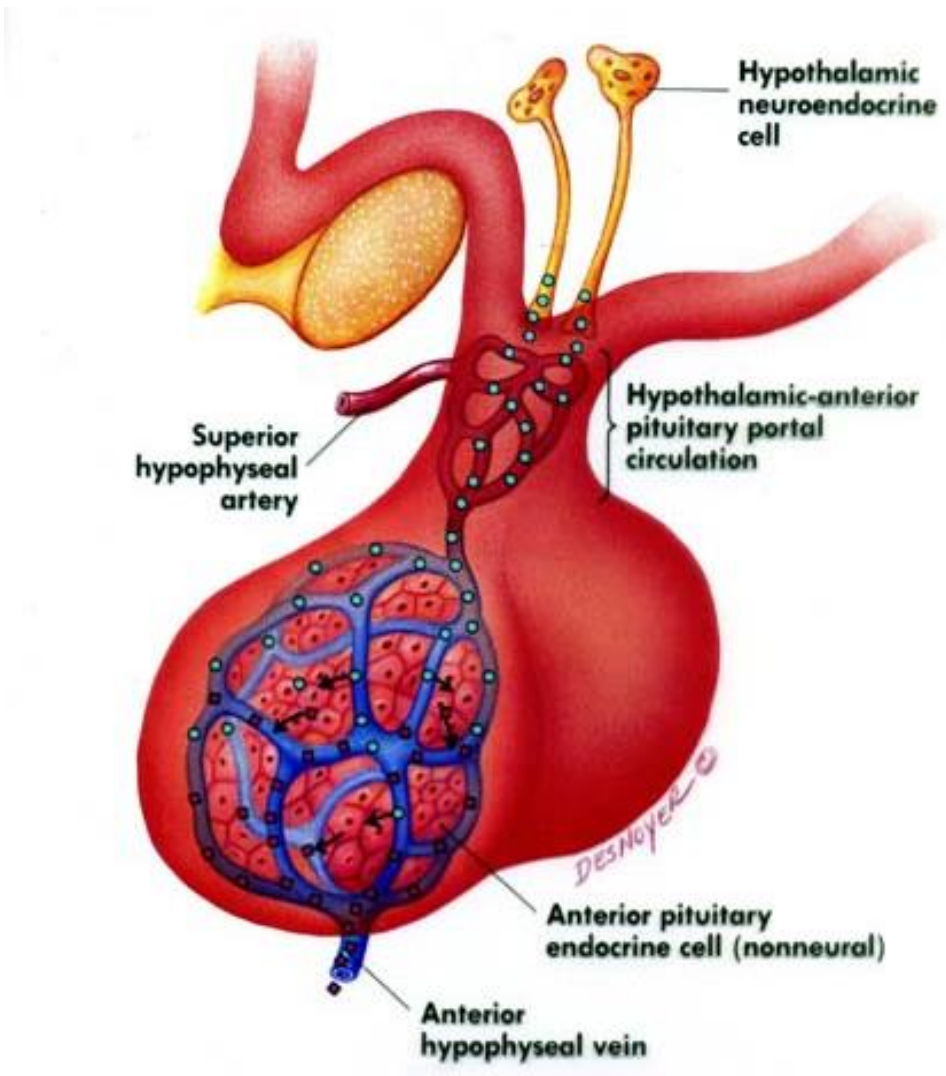
7. Describe the hormones, target organs, functions, and feedback regulation of the hypothalamic-pituitary-gonadal axis and hypothalamic-pituitary-breast axis. PHYS
8. Explain the pharmacologic treatment of prolactinomas, including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM
9. Describe the action and release of the posterior pituitary hormones oxytocin and anti-diuretic hormone. PHYS
10. Explain the pharmacological treatment of ADH deficiency and central diabetes insipidus, including the mechanisms of action, key adverse effects, and monitoring of agents used in the medical management of these disorders. PHARM

Basic schematic of H-P-end organ axis



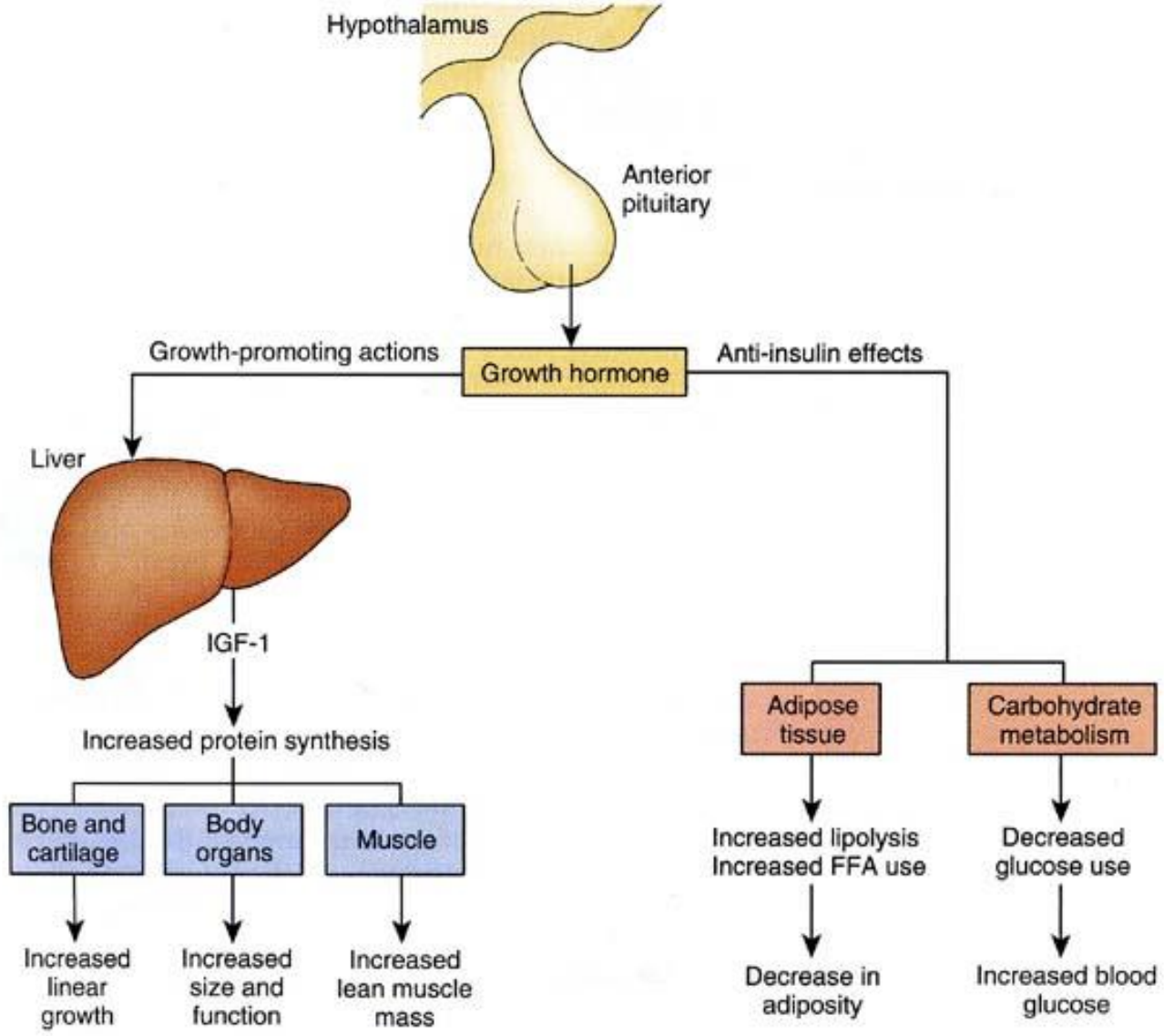

Parvocellular Neurosecretory Cells
Hypothalamus
Hypothalamic tropic hormone
Hypothalamic-pituitary portal vein
Anterior pituitary
Anterior pituitary tropic hormones
Systemic circulation
PRH
PIH (dopamine)
TRH
CRH
GHRH
GHIH (somatostatin)
GnRH
+
−
+
+
+
−
+
Endocrine cells
Prolactin
TSH
ACTH
GH
LH
FSH
Breasts
Thyroid gland
Adrenal cortex
Liver
Cells throughout body
Gonads
Male
Female

Structure of anterior pituitary

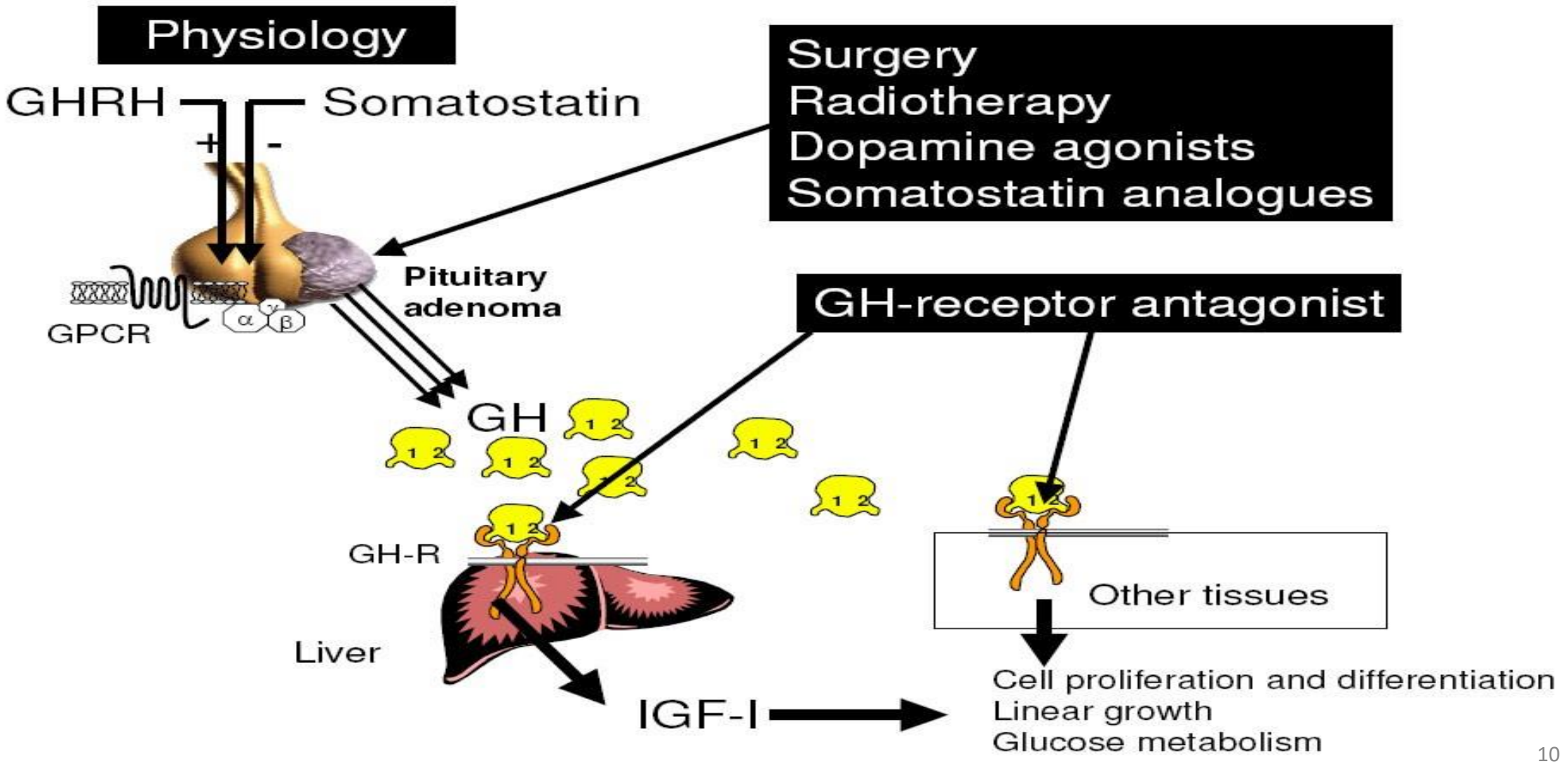


H-P-Growth Hormone axis revisited

Str

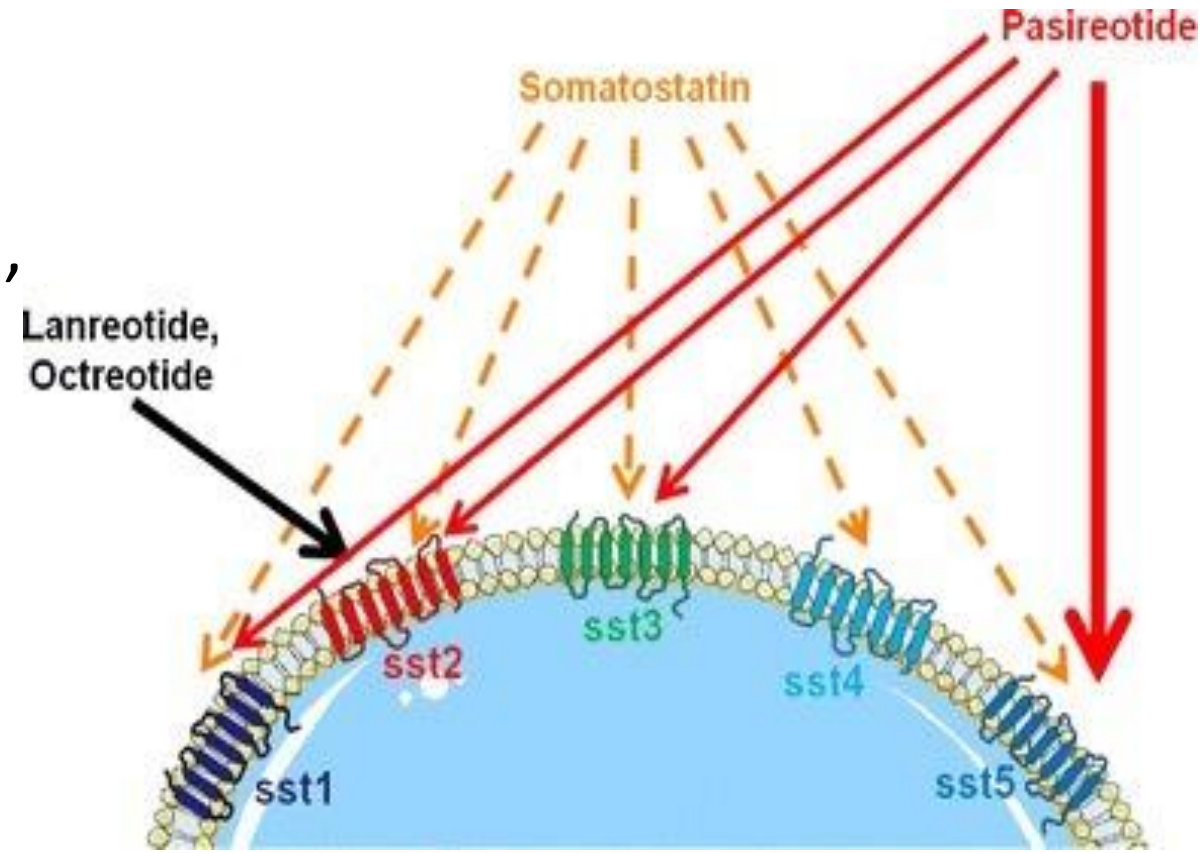


Acromegaly – targets for therapy



Available Somatostatin Analogs

- Octreotide – acts on SSt2
- Lanreotide – acts on SSt2
- Pasireotide – acts on SSt1, SSt2, SSt3 and sst5)



Side effects

- Abdominal cramps, flatulence, and diarrhea are common
- Occasional local skin irritation and pain at the injection site.
- Less common are reversible hair loss and, rarely, alopecia.
- Glucose control usually improves but rarely may worsen
- Gallbladder stones and sludge occur in approximately 25% of subjects and are usually asymptomatic.

Dopamine Agonists

- Dopamine agonists may inhibit GH secretion in some patients with acromegaly but do not work as well as somatostatin analogs.

- Cabergoline is the most effective dopamine agonist for the adjuvant management of acromegaly and is therefore the drug of choice in this category.

Pegvisomant

- GH receptor antagonist
- Competes with endogenous GH for binding at its receptor and blocks peripheral production of IGF-1
- Does not target the GH-secreting pituitary tumor
- Administered sc as 10-, 15-, or 20-mg daily injections
- Dose-dependent normalization of IGF-1 levels was achieved in up to 63 % to 95% of patients receiving up to 40 mg daily

GHRH – GH deficiency

- Growth hormone deficiency (GHD) is effectively excluded in children with a normal bone age and height velocity. In this case, provocative testing for GHD is not required.

- IGF-I is the mediator of the majority of the growth-promoting actions of GH.

- Patients with suspected GHD should be screened with **insulin-like growth factor I (IGF-I)** and serum levels of insulin-like growth factor binding protein-3 (IGFBP-3 – used more in pediatrics).

GH deficiency treatment

- Recombinant human GH (rhGH) is used to treat GH deficiency.

Administration and dosing: rhGH is administered by subcutaneous injection once a day, usually in the evening. Usually started at a lower than maintenance dose and is gradually increased.

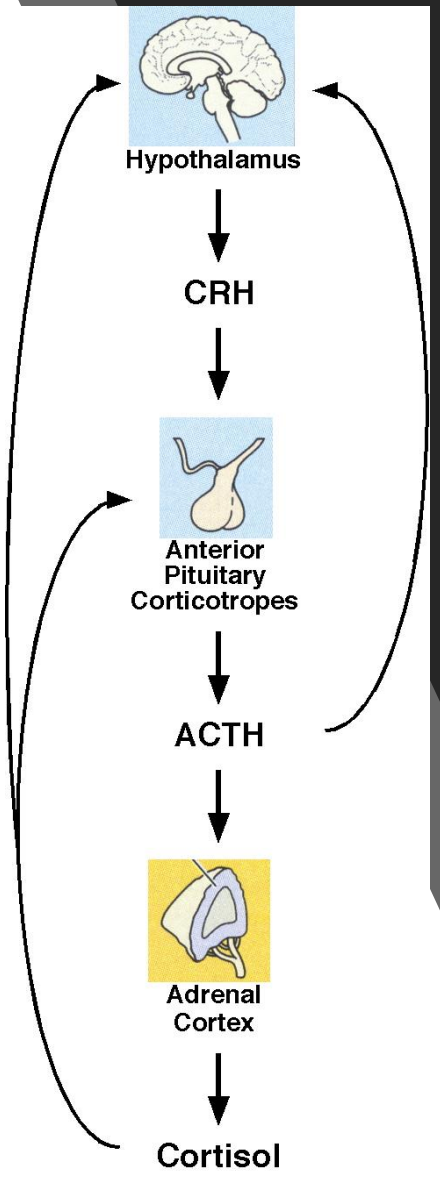
Goals of therapy: maintain the serum IGF-1 concentration within the age adjusted mid normal range.

Monitoring: Measurement of serum IGF-1 is the best single test to assess the adequacy of GH treatment.

In pediatric patients, monitoring growth response (i.e. height or height velocity) is also important.

GH treatment – side effects

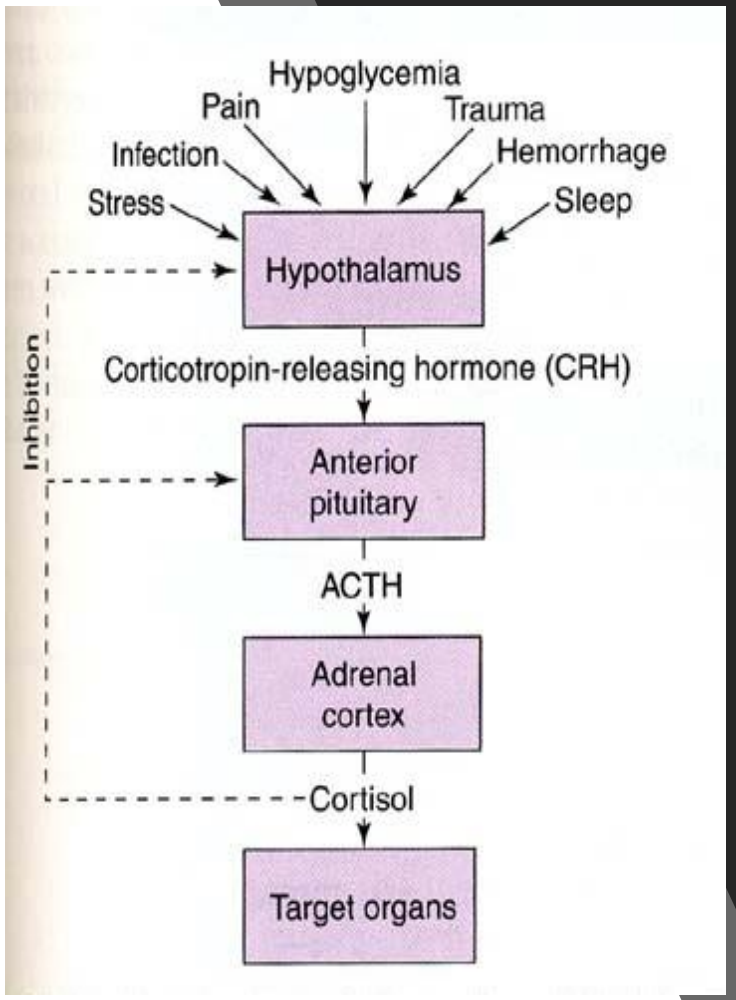
- peripheral edema,
- headaches,
- arthralgias,
- carpal tunnel syndrome,
- paresthesias
- worsening of glucose tolerance.
- **Contraindication:** Active malignancy is considered to be a contraindication to GH treatment because of the theoretical possibility that the treatment could stimulate tumor growth, but available evidence to date does not support this concern.



Structure of H-P-adrenal axis

- **Hypothalamus & Corticotropin Releasing Hormone (CRH)**
- **Anterior pituitary & Adenocorticotropin Hormone (ACTH)**
- **Adrenal cortex & Cortisol**
- **Cortisol, its target cells & feedback loop**

Inputs to hypothalamus affecting adrenal function



From Porth, 2005

- Stress
- Infection
- Pain
- Hypoglycemia
- Trauma
- Hemorrhage
- Sleep

Circadian patterns

- **Most effective stimulus for H-P-adrenal stimulation is the circadian rhythm**
- **CRH is released at or around awakening**
- **Circadian pattern involves a 24-hour cyclic pattern**
- **Regulated by melatonin**
- **Tied to dark and light cycles**
- **Can be disrupted as occurs in jet lag and shift work**

Normal Circadian Pattern

- **For the adrenocortical axis:**
  - **Peak hormone levels between 5 and 9 am**
  - **Low levels in the evening hours**

- **This has significance in overnight glucose levels**

Additional stimuli to CRH

- **Even if chemical neurotransmission & circadian patterns are normal, stress stimuli can override the normal pattern**
- **Anxiety, pain, surgery, chemical intoxication, metabolic causes (such as hypoglycemia)**

Corticotropin releasing hormone

- **from the hypothalamus, it enters portal circulation at the median eminence**
- **travels to the anterior pituitary where it causes the synthesis and secretion of ACTH**
- **its structure is that of a 41 amino acid peptide**

Anterior pituitary

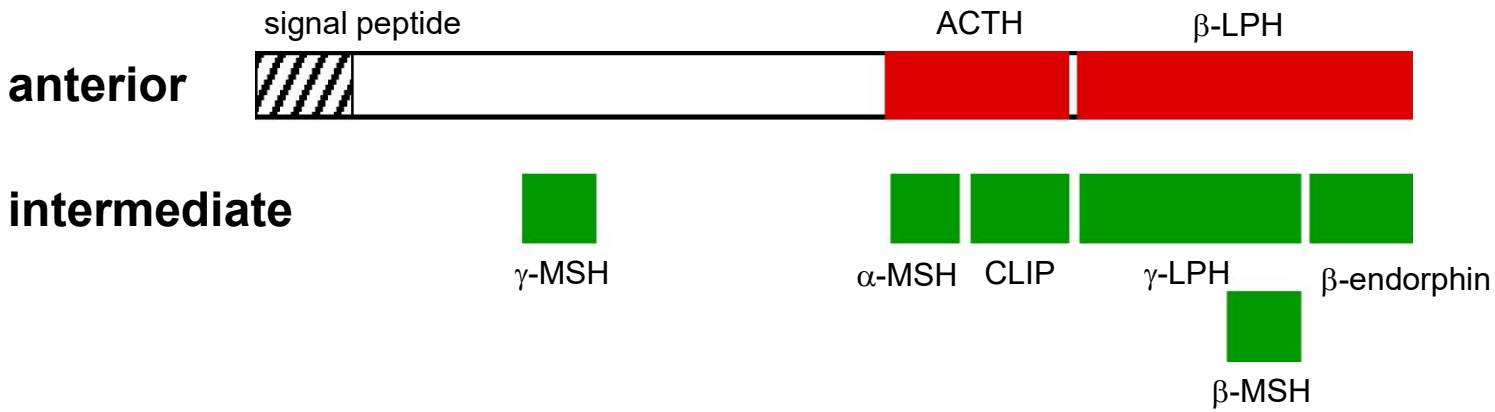
- **ACTH is produced and released by the corticotrophs of the pituitary**
- **ACTH is formed from a precursor molecule called pro-opiomelanocortin**

Pro-opiomelanocortin

- **can be broken down into:**
  - **N terminal segment**
  - **ACTH which includes melanocyte stimulating hormone**
  - **beta-LPH (lipotropin) which includes endorphins, enkephalins and others**

Polyprotein: pro-opiomelanocortin (POMC)
single precursor - multiple hormone products

Pituitary lobe:



ACTH

- 39 amino acid molecule
- first 18 amino acids essential for its biological activity
- secretion is episodic
- it is circadian:
  - 150 pg/ml around 8:00 am
  - <0-35 pg/ml at midnight
- target cells are cells of adrenal cortex

Functions of ACTH

- maintenance of adrenal gland size
- activation of cAMP system
- mobilization of cholesterol
- conversion of cholesterol to pregnenolone
- synthesis and secretion of steroids

Functions of ACTH

- **through all of its direct functions ACTH directs production of:**
  - **cortisol**
    - **main glucocorticoid hormone**
    - **main feedback hormone on corticotropes**
  - **androgens**
  - **indirect/lesser effect on production of aldosterone**

ACTH excess – pituitary Cushing (disease)

- Primary therapy consists of transsphenoidal surgery
- Medical therapy is considered if surgery is delayed, contraindicated, or unsuccessful
- Medical management is also used pre-op to stabilize patients with very high cortisol levels
- Remember, untreated Cushing disease can lead to uncontrolled hypertension, hyperglycemia, hypokalemia, myopathy, psychosis

Available medical treatments

1. Somatostatin analogs

2. Ketoconazole

3. Metyrapone

4. Mitotane

5. Mifepristone

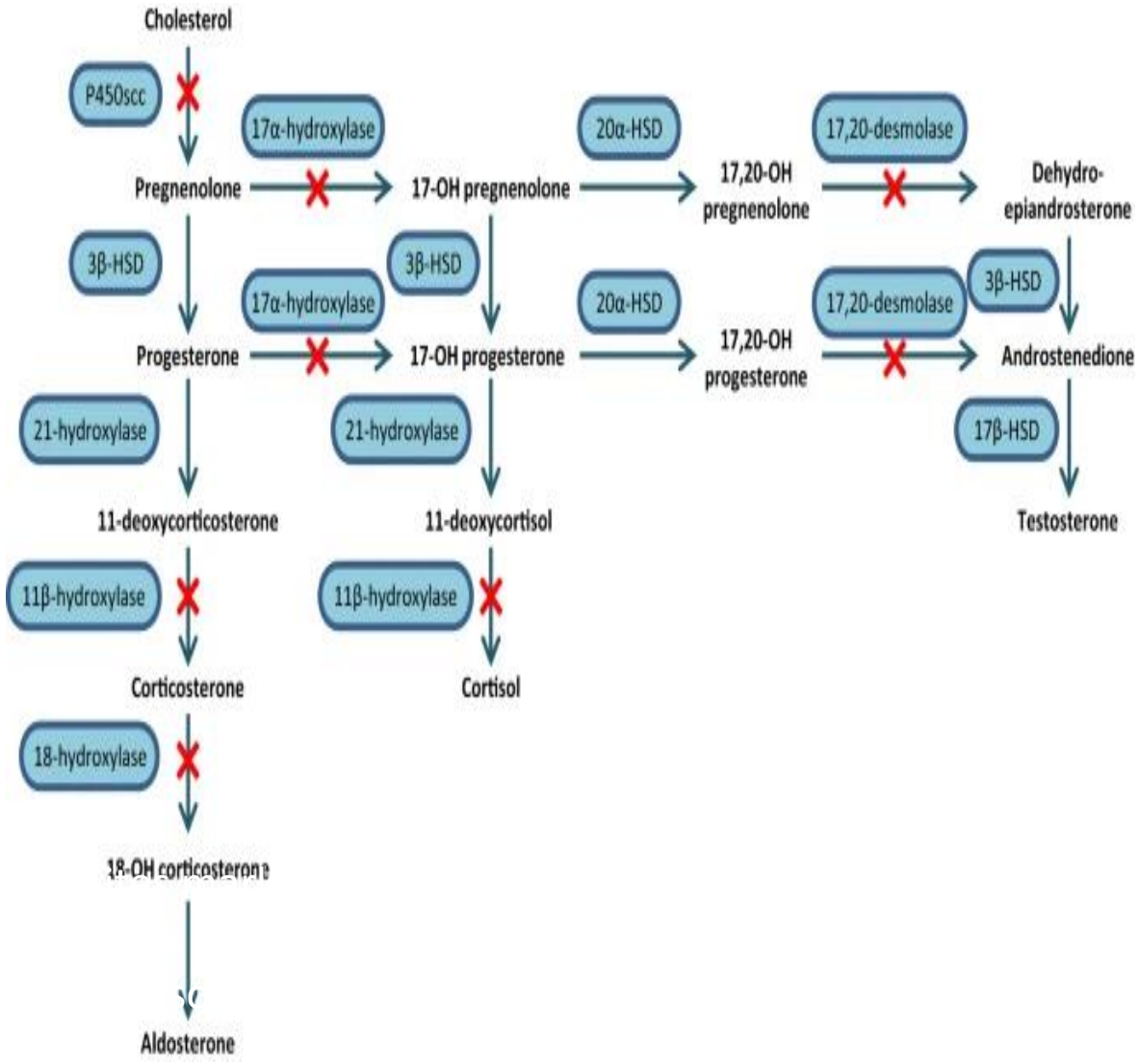
Somatostatin analog - Pasireotide

- Binds to somatostatin receptors and blocks the release of corticotropin (ACTH) from the corticotropes via its high affinity for somatostatin receptor subtype 5.

- Reduce and normalize urinary free cortisol and improve clinical symptoms.

Ketoconazole: inhibits adrenal cortisol synthesis

- Inhibits the first step in cortisol biosynthesis (side-chain cleavage) and, to a lesser extent, the conversion of 11-deoxycortisol to cortisol.
- The side effects include headache, sedation, nausea, and vomiting.
- In biologic males, inhibition of androgen production may lead to gynecomastia, decreased libido, and impotence.
- Reversible hepatotoxicity.



Other agents

- Metyropone is a 11-beta-hydroxylase (P450c11) inhibitor that blocks the final step in cortisol biosynthesis.

- Mitotane is an adrenocorticolytic drug that is used primarily for the treatment of adrenal carcinoma. It can also be used to achieve medical adrenalectomy with or without pituitary irradiation in patients with Cushing's disease.

Glucocorticoid receptor antagonist: Mifepristone

- Mifepristone (RU-486) is an anti-progestational drug that is best known as an abortifacient. However at much higher doses, it acts as a **glucocorticoid receptor antagonist.**

- Approved as a once-daily oral medication to control hyperglycemia secondary to hypercortisolism in adults with endogenous Cushing's syndrome and type 2 diabetes or glucose intolerance who have failed surgery or are not candidates for surgery.

Diagnosis of secondary (central) adrenal insufficiency

- Baseline cortisol , done 8 am

  less than 3 ug/dl highly suggestive

  greater than 11 ug/dl , makes diagnosis unlikely

- Stimulation testing

  cosyntropin , synthetic ACTH (1-24) , which has the biologic potency of native ACTH (1-39)

  peak post cortisol of 18 ug/dl or greater is normal

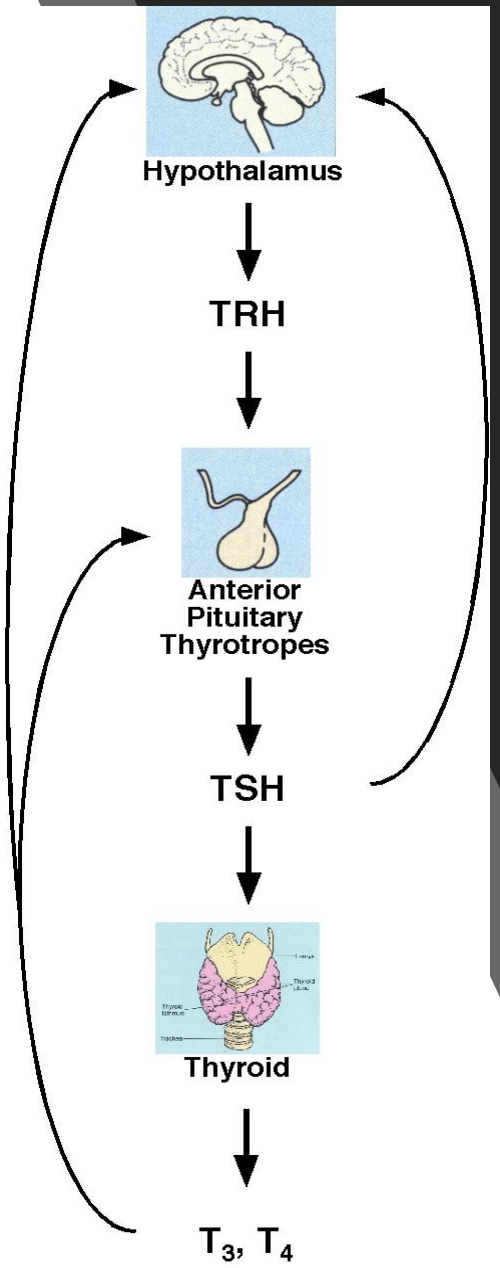
ACTH level

- Measure with am cortisol
- ACTH is low or inappropriately normal in central (secondary) adrenal insufficiency
- ACTH is high in primary adrenal insufficiency (Addison's)

Treatment of secondary (central) adrenal insufficiency

- Hydrocortisone 20mg in divided doses (15mg in AM, 5 mg late afternoon)
- Prednisone 5mg every AM
- Dexamethasone 0.5mg every AM

- In general, no need for mineralocorticoid (fludrocortisone) replacement since this is under control of the renin angiotensin system



Structure of H-P-thyroid axis

- Hypothalamus & thyrotropin releasing hormone (TRH)
- Anterior pituitary & thyroid stimulating hormone (TSH)
- Thyroid and thyroid hormone
- Thyroid hormone target cells & feedback loop

Thyroid axis hormones

- **thyrotropin releasing hormone (TRH)**
- **thyroid stimulating hormone (TSH, also known as thyrotropin)**
- **thyroxine (T_4) and triiodothyronine (T_3)**

Thyrotropin releasing hormone (TRH)

- **action:**
  - **induces quick release of TSH**
  - **effect peaks in 10-20 minutes**

TSH Secretion

- **Episodic and pulsatile**
- **Circadian rhythmicity**
- **Onset of sleep induces secretion**
- **If onset delayed get increases in release**
- **Relatively long half life allows effective laboratory measurement any time of day**

TSH Actions

- Cells of thyroid gland are target
- TSH binds to receptor and activates via cAMP
- Increases vascular flow through thyroid gland
- Increases uptake of iodide

TSH Actions

- Increases thyroid cell production of hormones
- Increases release of T_3 and T_4
- Acts as trophic factor on thyroid
- Chronic stimulation can produce hyperplasia and goiter

Treatment of secondary hyperthyroidism

Thyroid-stimulating hormone **(TSH)-secreting pituitary adenomas** are a rare cause of hyperthyroidism. They account for 0.5 to 3 percent of all functioning pituitary tumors and much less than 1 percent of all cases of hyperthyroidism.

Labs generally show elevated TSH with an elevated free T4 and or free T3.

Treatment is **surgery**. Medical therapy is used to treat patients who are not candidates for surgery. It can also be used to restore euthyroidism prior to surgery or for patients that have residual disease after surgery.

For medical therapy, most experts recommend a long-acting somatostatin analog (**octreotide**).

Diagnosis of hypothyroidism

Central (secondary) hypothyroidism

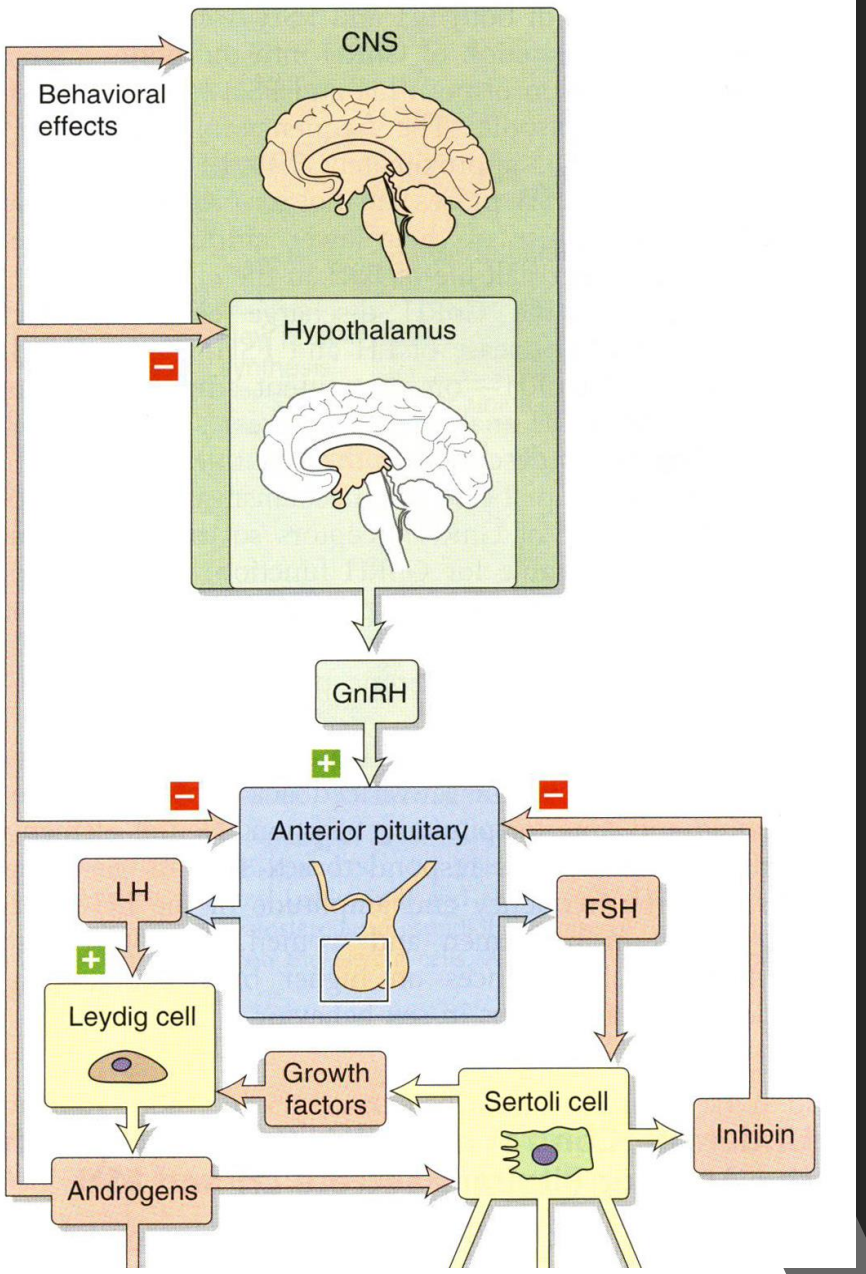
- Low free T4
- Low, normal to mildly (inappropriate) elevated TSH

Primary hypothyroidism

- Low or normal free T4
- Elevated TSH

Treatment of secondary hypothyroidism

- Synthetic T4, levothyroxine
- Dose of 1.6 ug/kg daily, start at a lower dose in those with cardiovascular disease and elderly
- Goal is to target free T4 in normal range
- DO NOT TARGET therapy with TSH levels



Structure of H-P-gonadal axis - testicular

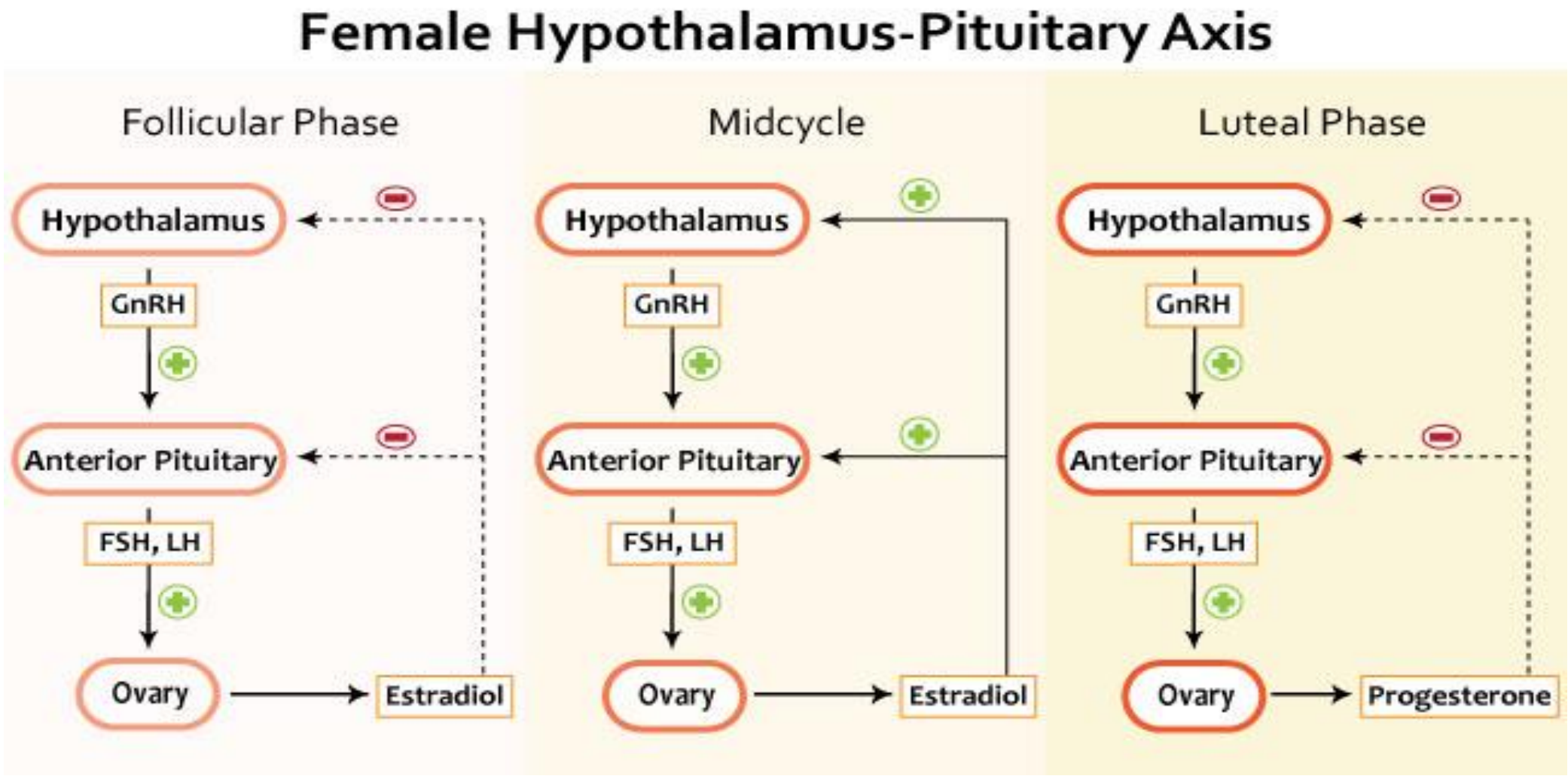
Hypothalamus & gonadotropin releasing hormone (GnRH)

Anterior pituitary; luteinizing hormone (LH) and follicule stimulating hormone (FSH) – most gonadotropes produce both

Leydig and Sertoli cells of testes – stimulate sex hormone synthesis (steroidogenesis) and spermatogenesis

Androgens, target cells & feedback loop

Structure of H-P-gonadal axis - ovarian



H-P-gonadal axis - ovarian

- GnRH → ↑ FSH and ↑ LH (pulsatile)
- LH → (+) theca cells → ↑ progesterone and androstenedione
- FSH acts on granulosa cells
- Negative feedback predominates, except in midcycle, when positive feedback is in effect

H-P-gonadal axis - ovarian

- The pattern of GnRH changes during the menstrual cycle
- During the luteal to follicular transition, with low estradiol and low inhibin, GnRH pulses occur every 90-120 minutes, and FSH predominates
- During mid- to late-follicular phase, with increasing estradiol and inhibin A, GnRH pulses occur more rapidly, and LH predominates
- Post ovulation, with increased estradiol, inhibin A, and progesterone, GnRH pulsatility decreases, and FSH release decreases

Structure of H-P-breast axis

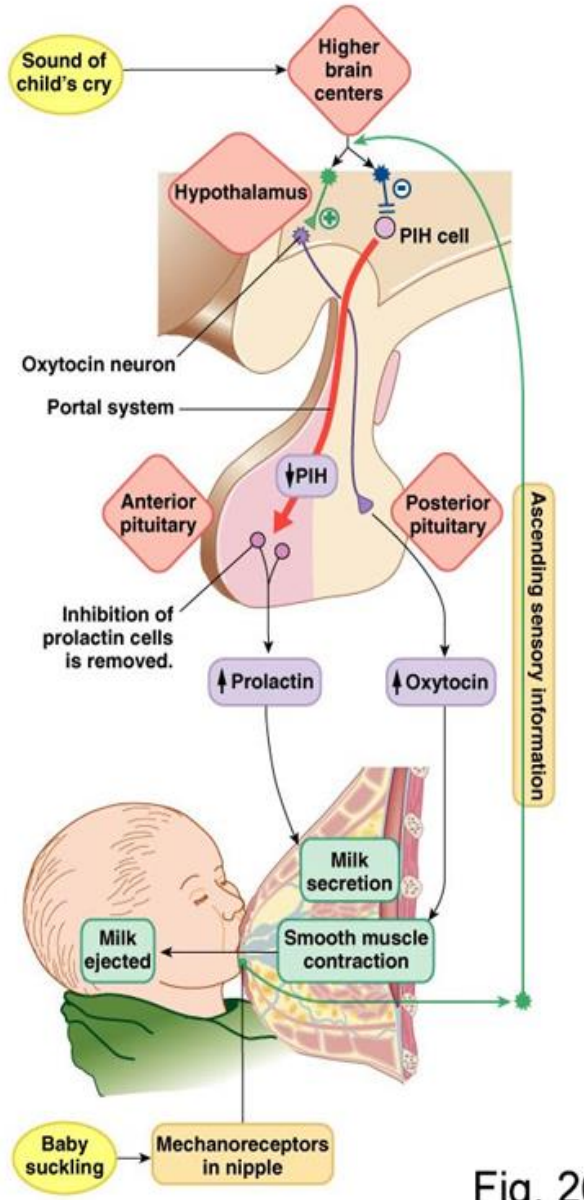


Fig. 26-23

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Prolactin is synthesized and secreted by lactotrophs

Lactotrophs are 15-20% of cells of anterior pituitary; however, this percentage increases with estrogen release, particularly with pregnancy

Prolactin release is predominantly under tonic inhibition by dopamine from hypothalamic dopaminergic neurons

GnRH also plays some role in prolactin release

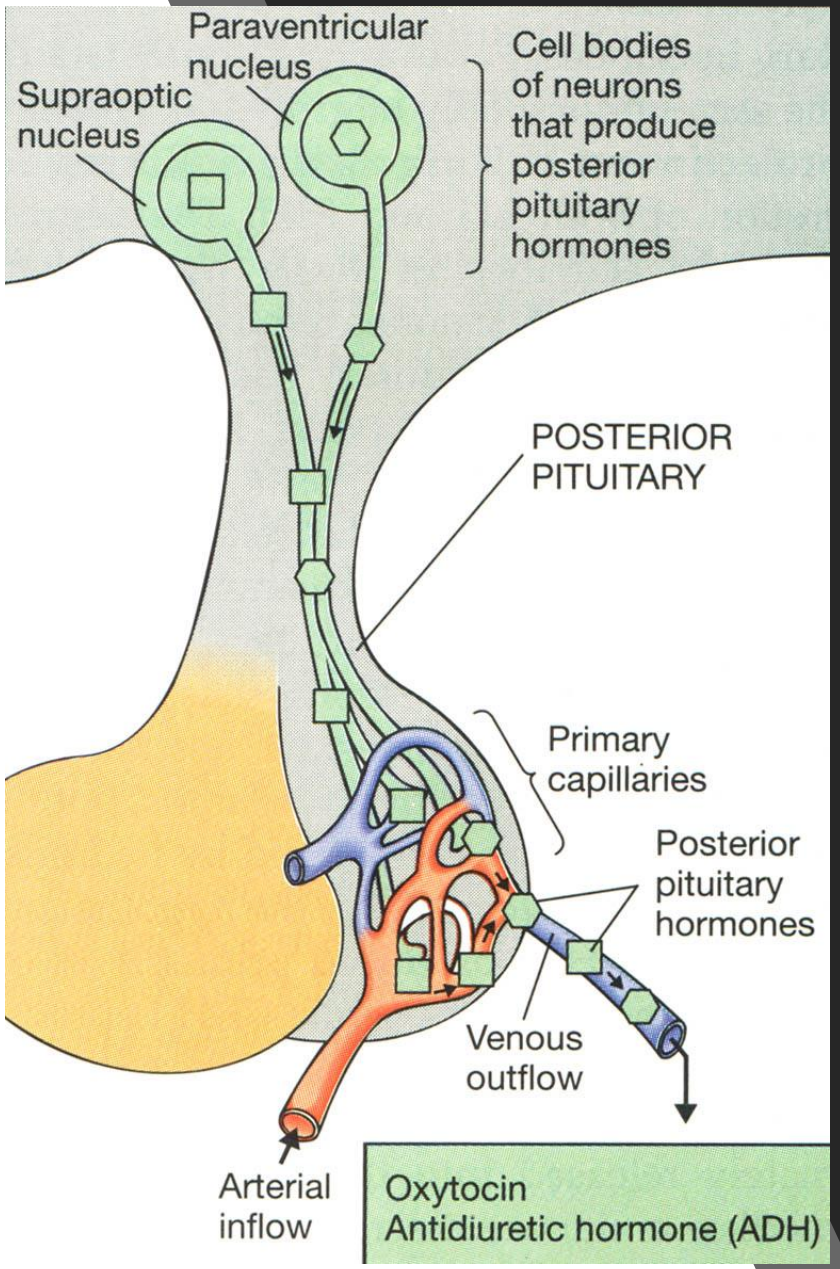
Prolactin release is increased by suckling. Suckling inhibits dopamine release, removing inhibition from the lactotroph

Medical Management of Prolactinomas

- Primarily medical management , even when patients present with visual field defects
- Dopamine agonists (bromocriptine and cabergoline) are first line
- These lead to tumor shrinkage , and with micro adenomas, treatment can be stopped in 1 to 2 years
- Surgery and radiation reserved for those who do not respond to medical management

Dopamine agonists and their side effects

- Bromocriptine and cabergoline
- Dopamine agonists can cause nausea, headache, light-headedness, orthostatic hypotension, and fatigue.
- Cabergoline treatment at high doses (used mostly for Parkinson's disease) is associated with higher risk of valvular heart disease, but not at the lower dose used for hyperprolactinemia.
- Cabergoline appears to cause nausea less often than bromocriptine.



Hormones of the Posterior Pituitary

Extension of the hypothalamus – contains axon terminals of neurons located in supraoptic and paraventricular nuclei

Neurons generate action potentials – produce membrane depolarization and exocytosis of contents of secretory vesicles

Oxytocin and anti-diuretic hormone enter systemic circulation through venous drainage of posterior pituitary into the intracavernous sinus and internal jugular vein

Antidiuretic hormone

- binds with the V_2 receptors on the cells of the collecting duct
- this initiates the insertion of the Aquaporin 2 water channels
- without ADH, these are stored in endosomes within the cells

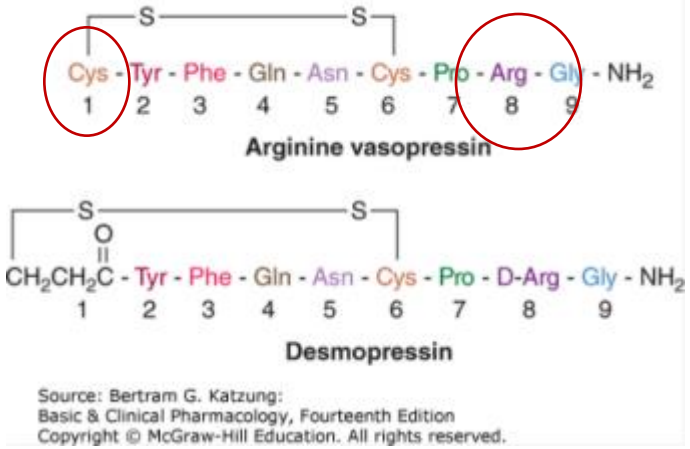
Antidiuretic hormone

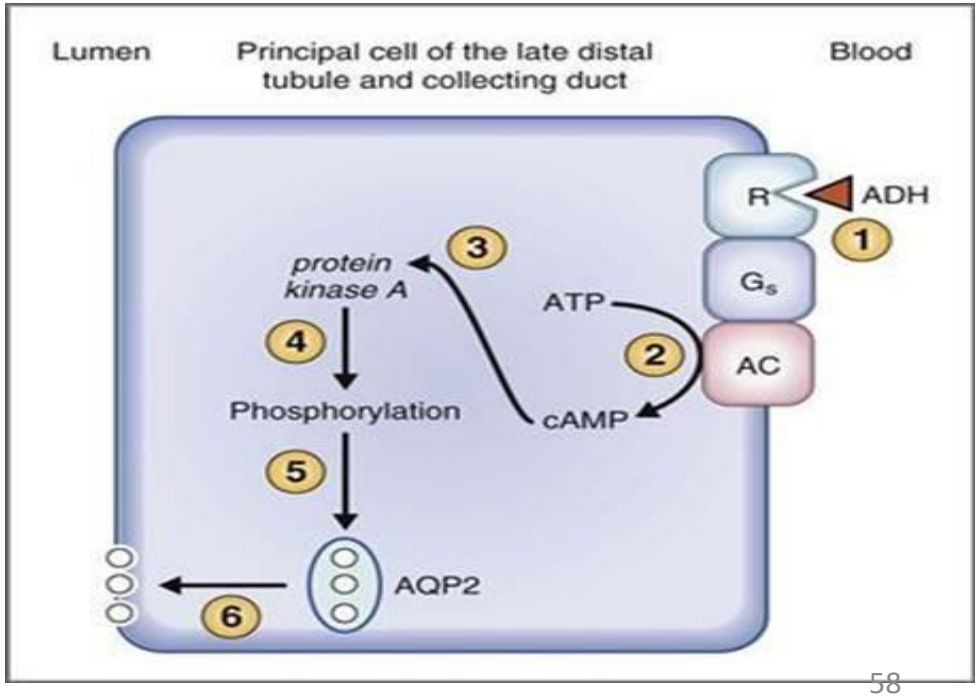
- with ADH, the cells of the collecting duct become permeable to water
- water is reabsorbed into medulary interstitium and eventually into blood
- urine becomes more concentrated as water is reabsorbed

Treatment of AVP deficient DI

- **<u>Desmopressin</u>** (dDAVP) is a two-amino acid substitute of ADH that has potent antidiuretic but no vasopressor activity.

- Acts on <u>V_2 receptors</u> in the distal tubules and collecting ducts → (cAMP) → increases water permeability.





DDAVP formulations

Available as an intranasal spray, tablet and parenteral formulation.

- Potency - Injectable > nasal spray > Oral.

- Injectable dosing (SQ) 2-4 mcg
- Nasal – 10-20 mcg
- Oral 100-200 mcg

Goals of therapy

The initial aim of therapy is to reduce nocturia, thereby permitting adequate sleep. Thus, the first dose is typically given at bedtime.

Maintain serum sodium in normal range

Major side effect - **hyponatremia.**

Monitoring: - Follow serum sodium. Hyponatremia can usually be avoided by giving the lowest dose possible.

Counsel patient about symptoms of hyponatremia - confusion, nausea, vomiting

Oxytocin

- Release of oxytocin is stimulated during lactation and childbirth
- In lactating breast, oxytocin stimulates milk ejection by stimulating contraction of cells lining alveoli and ducts in mammary gland
- In pregnant uterus, oxytocin produces rhythmic contractions to help induce labor and promote uterine regression after delivery
- At onset of labor, upregulation of oxytocin receptors in uterine muscle – up to 200 times that of non-pregnant uterus
- Contractile activity of uterus acts through positive feedback during parurition to stimulation oxytocin neurons
- Oxytocin may play a role in maternal behavior and amnesia in childbirth
- Increased levels during moments of intimacy – may breed trust
- Synthesized oxytocin is administered intravenously for initiation and augmentation of labor (**Pitocin**; you will hear it referred to as "pit" on the labor and delivery floor.)

Thank you.

Any questions?